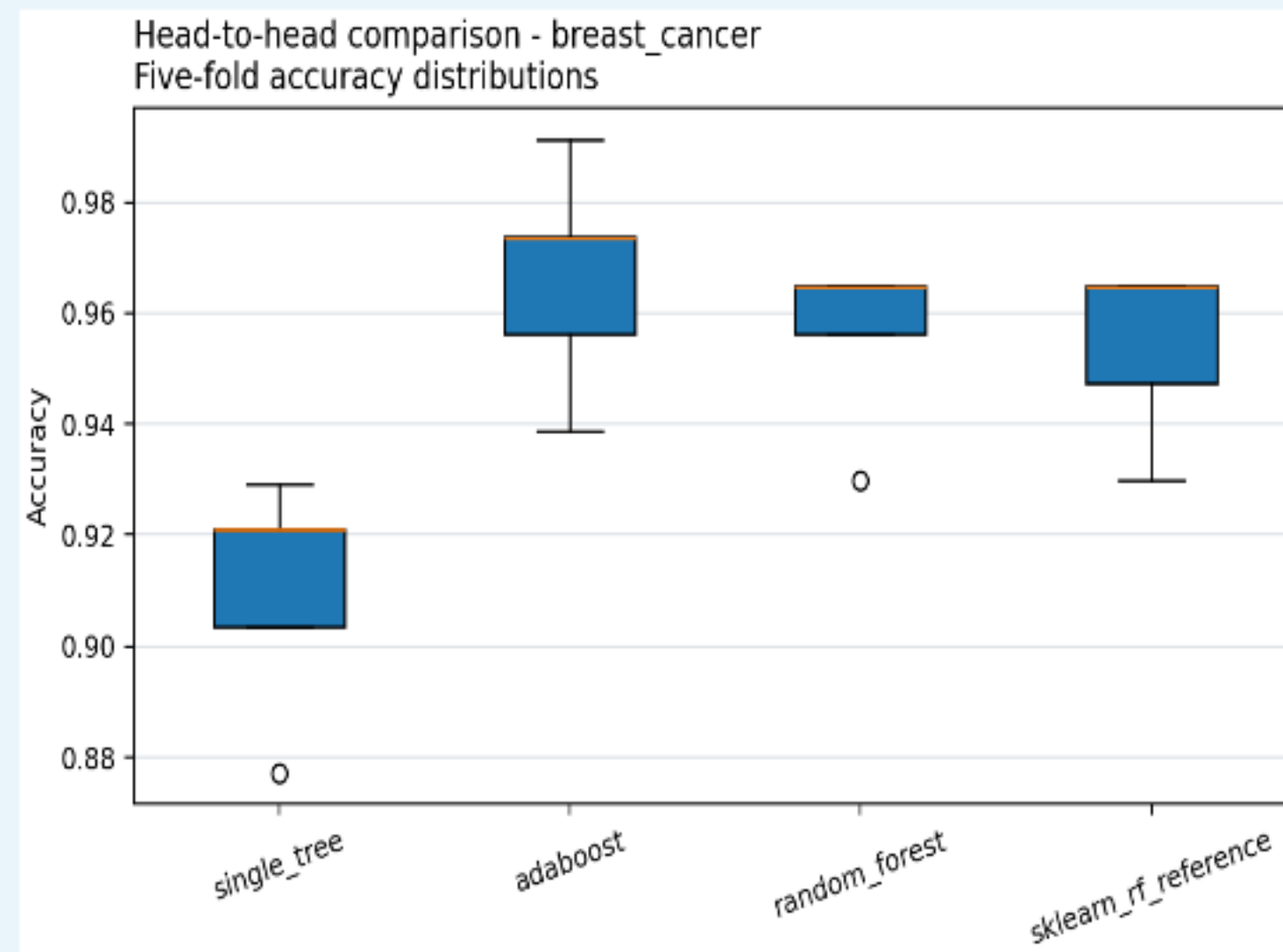


When does boosting beat bagging?

A from-scratch study of bias, variance, label noise, imbalance, and data structure

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The methods attack different errors

BOOSTING

Correct

Fit a stump

Raise hard-example weight

Sequential attention can lower residual boundary bias and can chase corrupted labels.

BAGGING

Average

Bootstrap

Decorrelate trees

Independent variation cancels; shared correlation remains. OOB rows provide an internal check.

Inspectable code - not a library wrapper

7

from-scratch algorithms

Tree | SAMME | Forest
PCA | K-Means | DBSCAN
Gradient Boosting

51

**passing tests; 88.71%
coverage**

Weighted splits | OOB
complements
Empty clusters | noise-to-border
Sequential/parallel identity

1

**binding reproduction
command**

python src/experiments/run_all.py
Seed 42 | manifests | raw
fold/stage/replicate evidence

Three datasets isolate the conditions that should change the winner

DATA

Breast Cancer

569 rows | 30 features | balanced
binary baseline and Brier study

Adult Income

48,842 rows | 14 raw features | mixed
types, missingness, moderate imbalance

Covertypes

10,000-row subset | 54 features | 7
classes | minority 0.47%

SEVEN REQUIRED EXPERIMENTS

1 Baseline and sklearn tolerance | 2 AdaBoost
stages: 1-200 | 3 Forest size and depth | 4
Identical five-fold head-to-head

5 Nested label-noise replicates | 6 100 shared
bootstrap fits | 7 PCA, K-Means, DBSCAN

Split first. Fit preprocessing and oversampling inside training only.

Custom trees stay within reference tolerance on every dataset

0.912

custom tree accuracy

0.912

sklearn tree accuracy

0.8 pp

maximum gap across datasets |
required ≤ 2.0 pp

Why the match is credible

Sorted midpoint search | weighted
impurity | deterministic ties | identical
train/test split

Why agreement is not identity

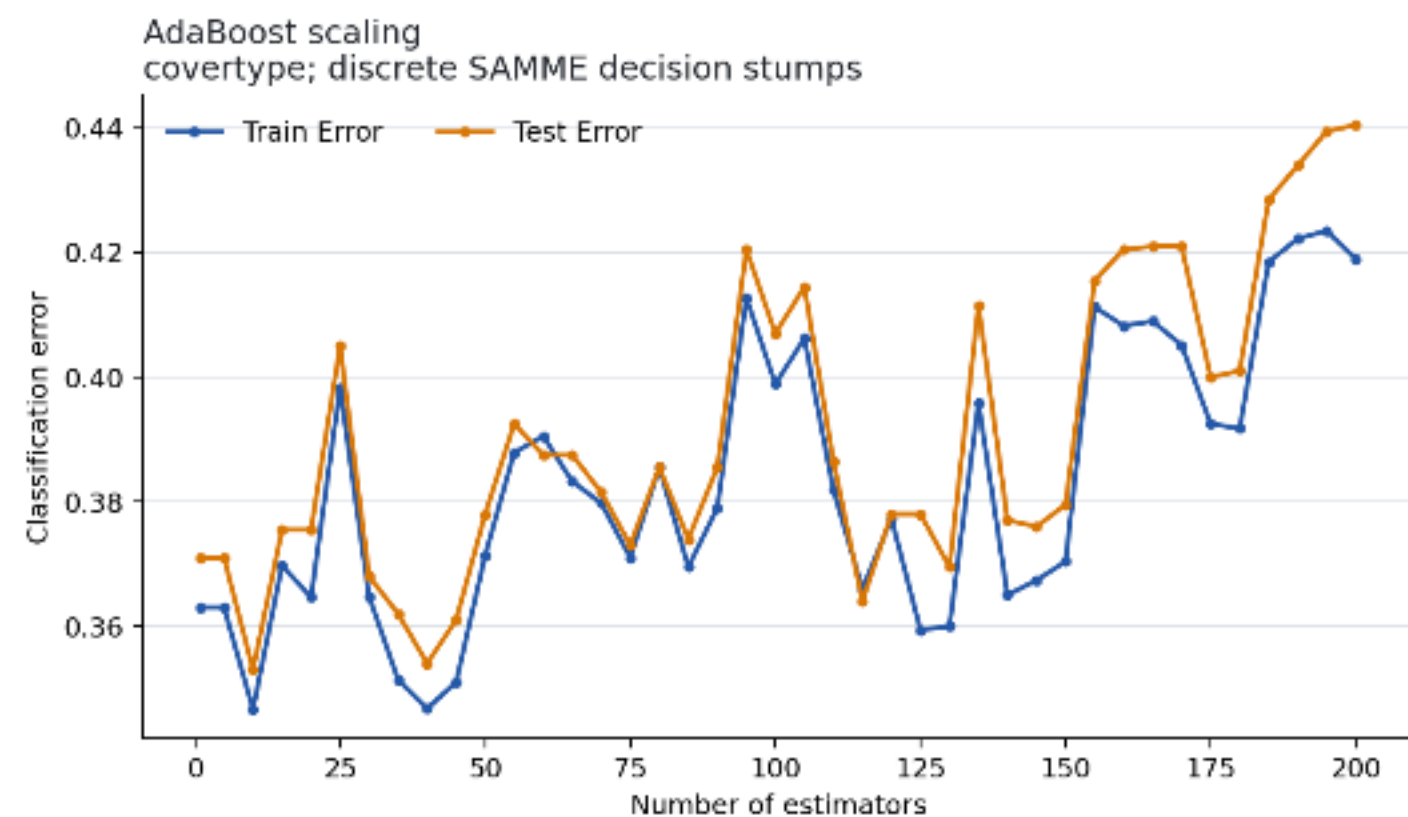
Different tie handling can change tree
shape; tests and invariants remain
necessary.

Acceptance

PASS

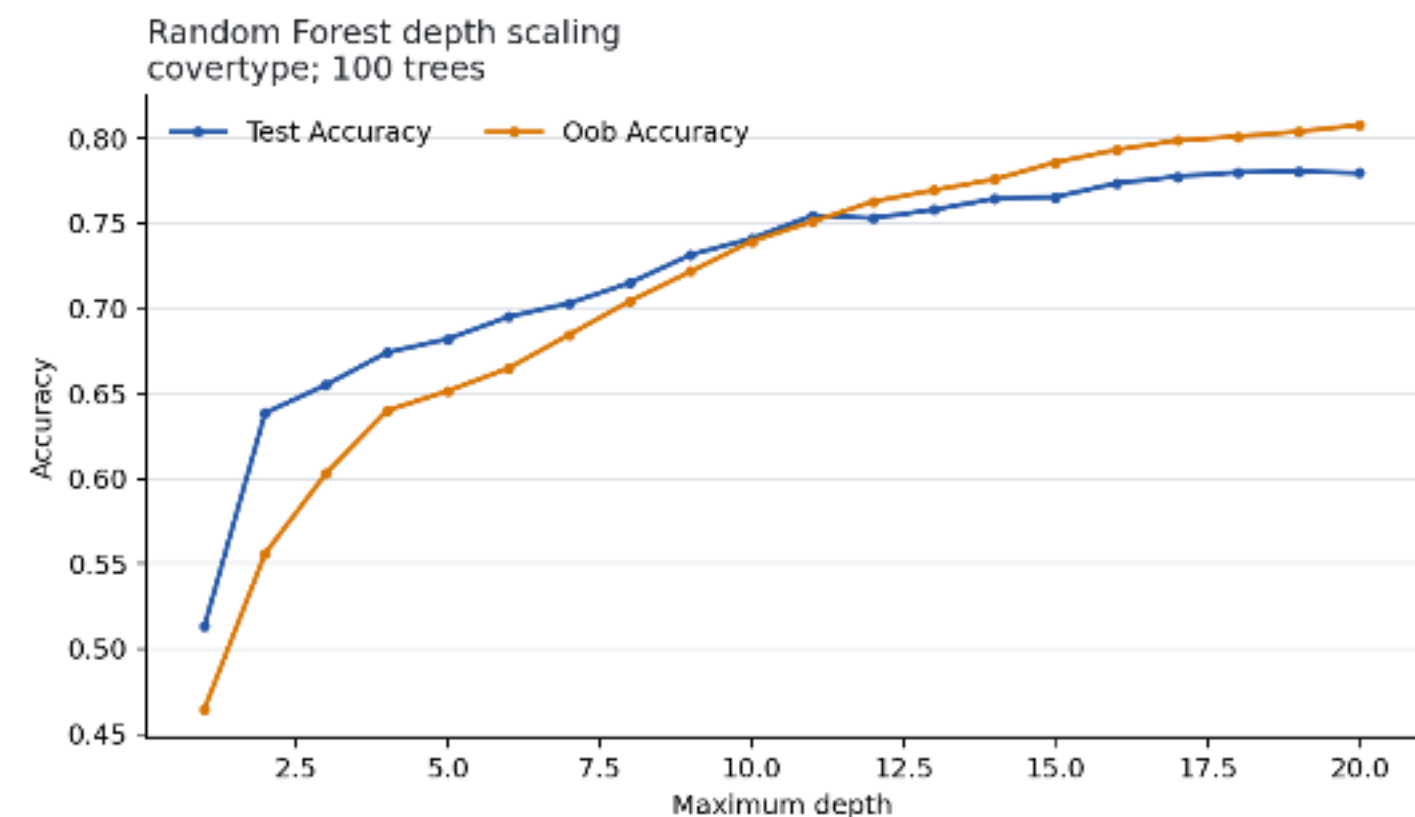
Scaling exposes different complexity needs

AdaBoost: 1-200



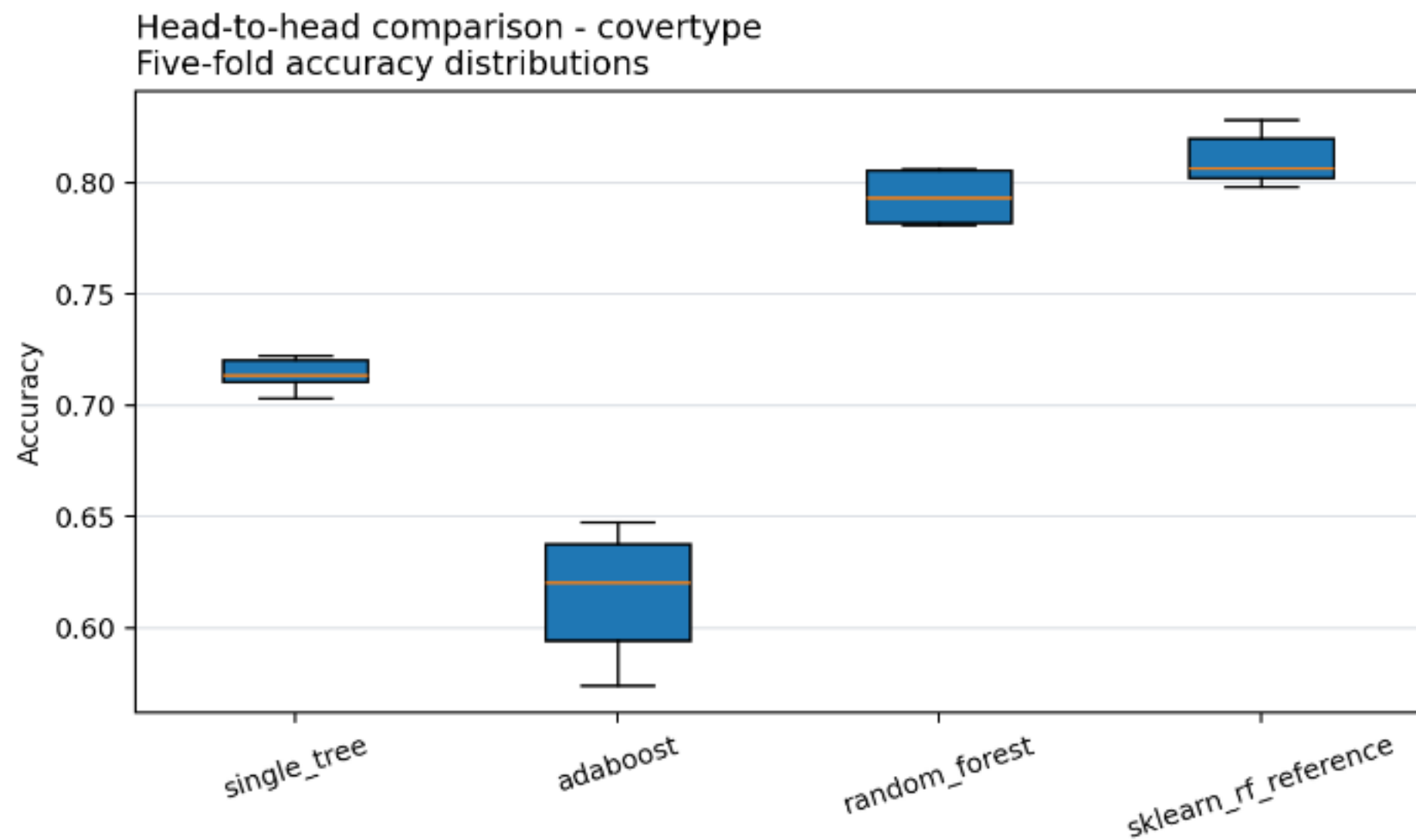
Best error at 10; later rounds hurt

Forest: depth 1-20



Best depth 19; flexible trees matter

The winner changes with dataset structure



FIVE-FOLD MEAN ACCURACY

.9666

Breast Cancer Ada | RF .9561

.8649

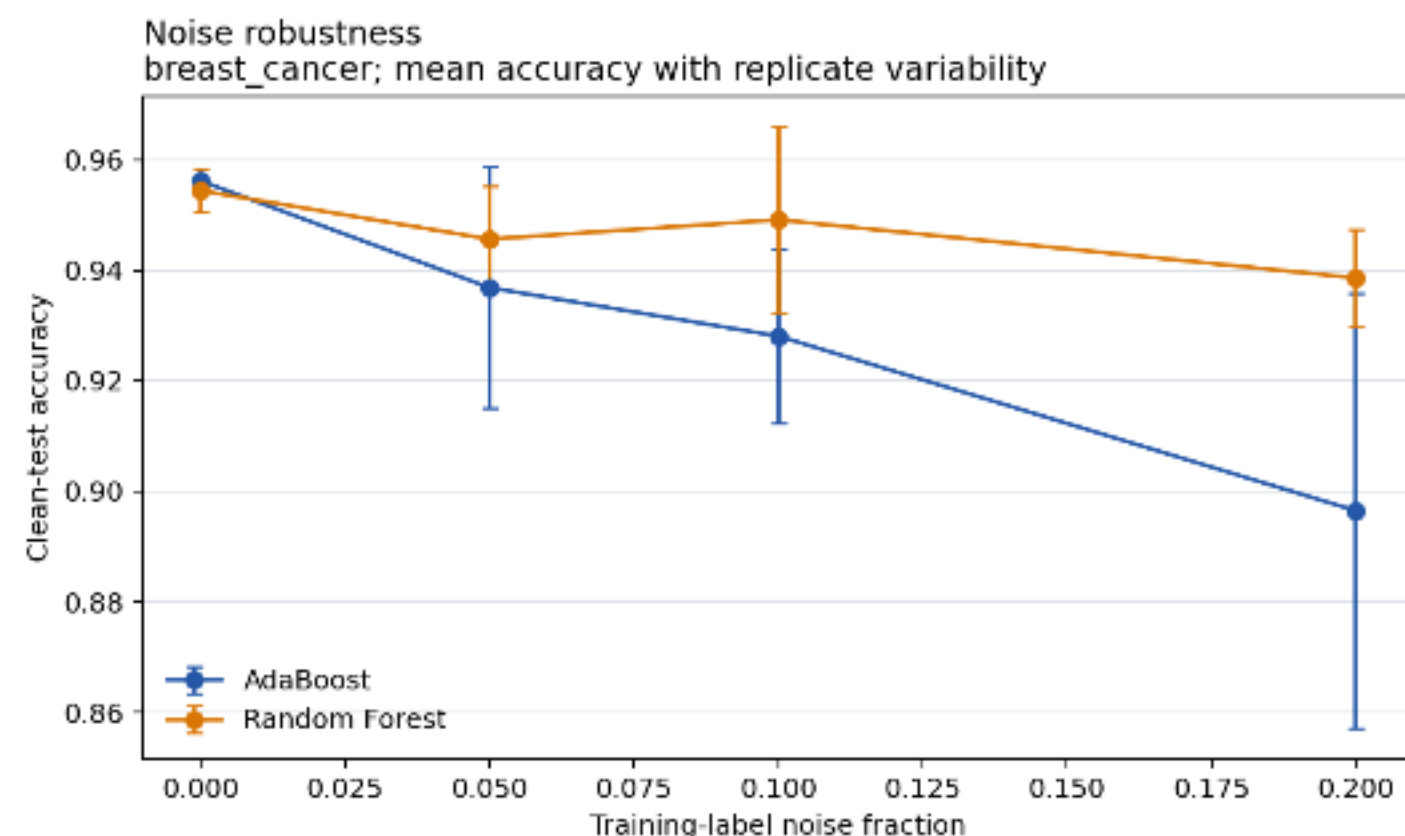
Adult RF | Ada .8556

.7938

Coverytype RF | Ada .6150

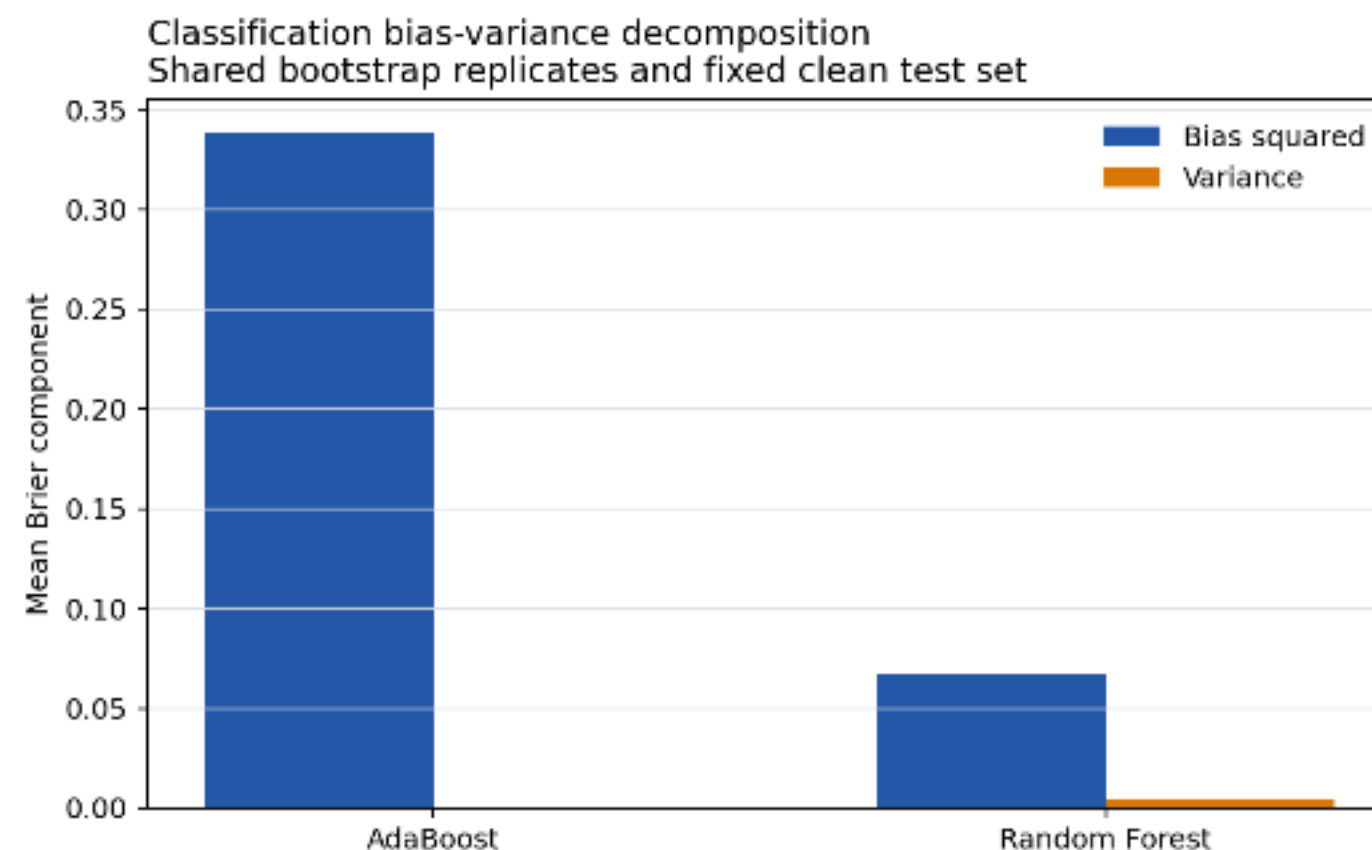
Noise robustness and probability loss tell different stories

20% LABEL NOISE | 5 REPLICATES



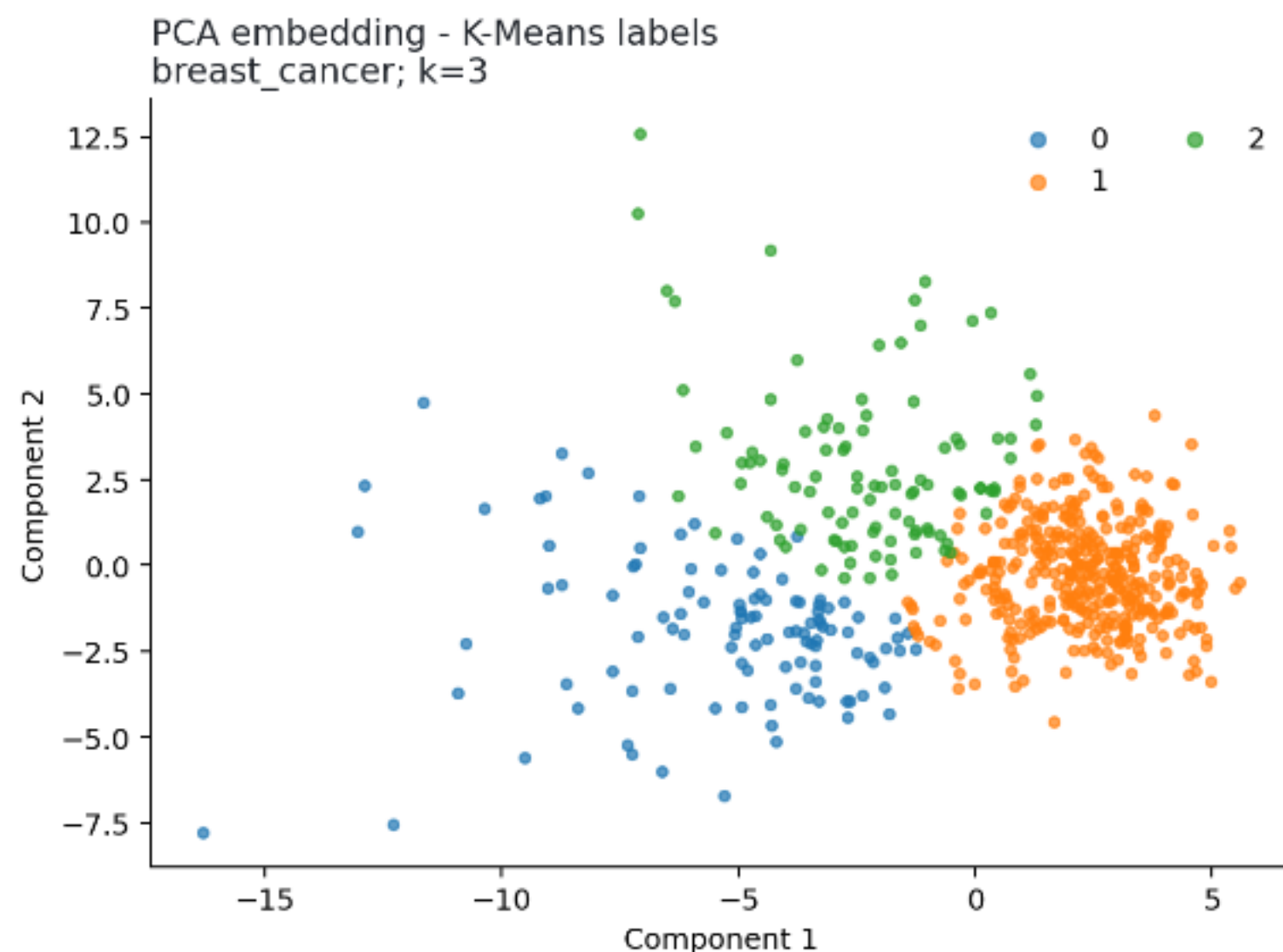
At 20%: AdaBoost -5.96 pp; forest -1.58 pp.

BRIER | 100 BOOTSTRAPS



Bias²: AdaBoost .3382; forest .0675. Votes are not calibrated.

Clusters reveal geometry, not guaranteed class boundaries



7

PCs reach $\geq 90\%$
variance

.515

K-Means ARI | k = 3

6.50%

DBSCAN noise |
ARI = .030

Full dataset n = 569

Bonuses extend the analysis without replacing it

SAMME.R

BC .9561 to .9649

Adult .8573 to .8687

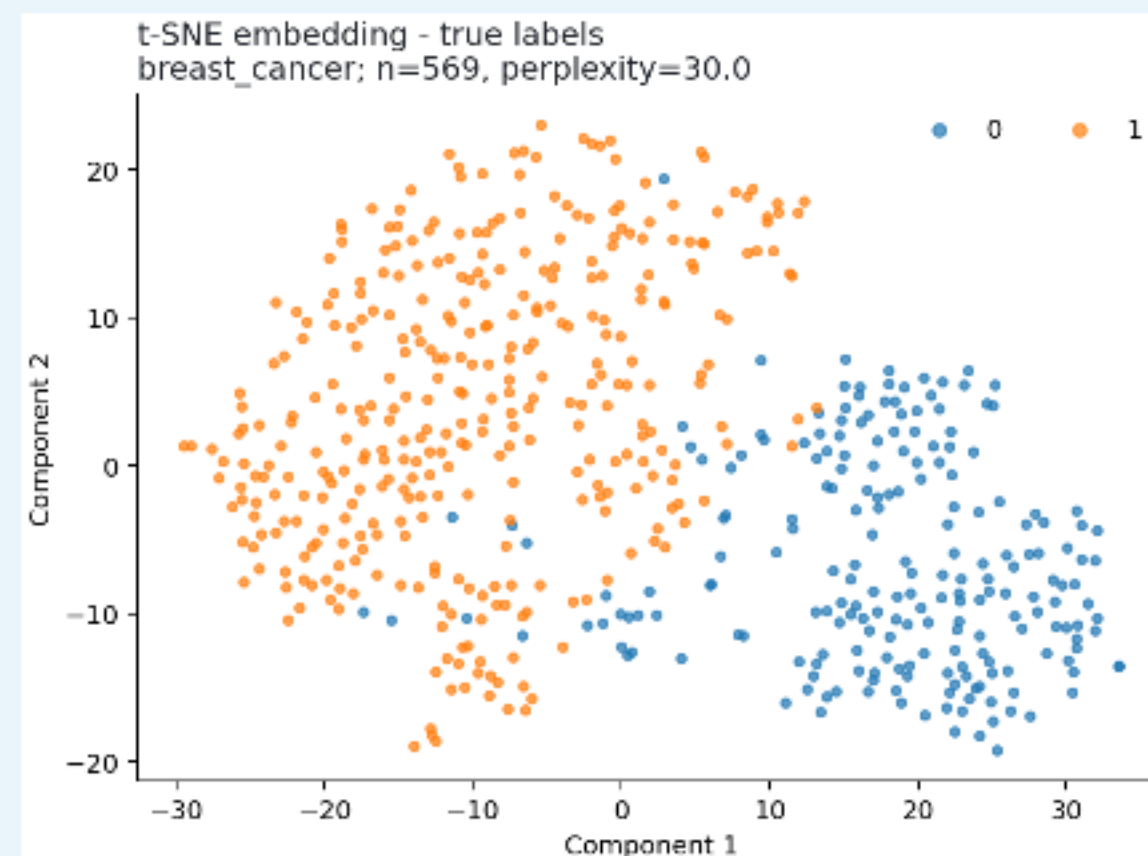
Gradient Boosting

Binary log-loss; BC accuracy .9386, AUC .9899

Interactive notebook

Executed controls for model and cluster complexity

t-SNE



n=569, perplexity 30 | no global-distance claim

No universal winner: let the data decide

BOOST

Breast Cancer: AdaBoost .9666. Sequential correction worked best on this structured binary task.

BAG

Adult RF .8649; Covertypes RF .7938. Diverse trees handled mixed and multiclass structure better.

Release after truthful contribution approval, genuine Git history, final audit, Moodle submission, and v1.0-final.